DDAD: Diffusion in Dimensionless Domain for Generalized Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) aims to distinguish background and anomalous targets in hyperspectral images (HSIs). In real-world applications, HSIs are often acquired under varying conditions, such as different sensors, times, and locations. This makes generalized HAD particularly important for practical deployment. Most existing methods follow an unsupervised reconstruction-based framework [e.g., autoencoders, generative adversarial networks (GANs)] that detects anomalies by measuring reconstruction errors. However, two spectral barriers remain: Spectral Divergence and Spectral Confusion, that hinder the realization of generalized HAD: 1) cross-scene HSIs exhibit diverse spectral characteristics. The spectral divergence in the original spectral space leads to degraded generalized detection performance and 2) most methods rely on pixel-level detection in the original space, which is prone to mixed-pixel effects that hinder accurate background reconstruction. To break these spectral barriers, we propose a novel dimensionless diffusion framework for generalized HAD (DDAD). First, we introduce a dimensionless domain transformation (DDT) module that maps HSIs from different sources into a unified, spectral-invariant dimensionless domain, which effectively eliminates spectral divergence and confusion across scenes. Next, we propose a controllable dimensionless diffusion reconstruction (CDDR) module that achieves balanced background reconstruction and anomaly suppression through controllable explicit background modeling. Additionally, to further enhance generalization, we design a dynamic spatial–spectral perturbation (DSSP) mechanism to mitigate the model's reliance on single-source characteristics. Extensive experiments on four benchmark datasets demonstrate the superior cross-scene generalized performance of DDAD.**

***Index Terms*—Anomaly detection, cross-scene hyperspectral images (HSIs), dimensionless diffusion model.**

I. Introduction

HYPERSPECTRAL images (HSIs), as one of the important remote sensing data, comprise hundreds of consecutive wavelength bands. HSIs acquired at different times, locations, or sensors often have radiometric, geometric, and spectral variations [1]. Hyperspectral anomaly detection (HAD) task aims to automatically identify anomalous targets of interest in HSIs without prior knowledge. In practical applications such as environmental monitoring, mineral exploration, and disaster assessment, there is increasing demand for generalized models that can adapt to diverse data sources and scenes. Consequently, the development of robust generalized HAD holds substantial practical value. Existing HAD methods can be categorized into two groups: traditional and deep learning-based methods. Traditional methods [2], [3], [4], [5] typically achieve reliable detection performance by accurately modeling the background, but require manual parameter tuning when applied to unseen scenes, hindering their generalization. Most of the previous deep learning-based methods [6], [7], [8], [9] are in the single-image training and detection paradigm, as shown in Fig. 1(a), which inherently lack cross-scene generalizability due to dependency on scene-specific spectral distributions. The few existing cross-scene methods [10], [11], [12], [13] are all performed on high-dimensional HSIs in the original spectral domain. As shown in Fig. 1(b), recent methods commonly utilize anomaly simulation strategies to generate a large number of sample pairs for learning the feature differences between background and anomalies. Moreover, the realization of generalization in existing methods overlooks two spectral barriers.

1) *Spectral Divergence Across Various Scenes:* HSIs acquired from different sources (e.g., sensors, locations, or times) often exhibit distinct spectral characteristics. When models are trained directly on the original spectral space, they inherently capture scene-specific features. This leads to poor generalization and degraded performance when applied to unseen target domains.
2) *Spectral Confusion of Pixel-Level Detection:* The limited spatial resolution of HSIs often leads to mixed pixels, where a single pixel represents a mixture of background and anomalies. This spectral confusion increases the label uncertainty and blurs the decision boundary, diminishing the discriminability between anomaly and background spectra. As a result, existing methods often struggle to accurately reconstruct complex background distributions across scenes.

To break the barriers of spectral divergence and spectral confusion, we propose a novel generalized HAD framework, dimensionless diffusion framework for generalized HAD (DDAD), based on diffusion reconstruction in the dimensionless domain. Specifically, we introduce the dimensionless

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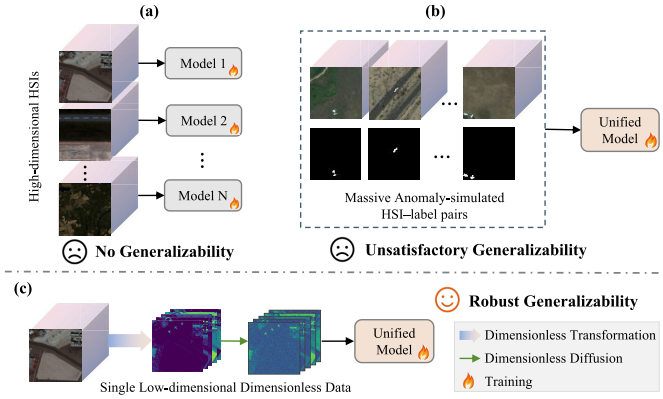


Fig. 1. (Top) Comparison between existing HAD frameworks and (bottom) our DDAD. Existing frameworks rely on the original high-dimensional spectral domain, susceptible to spectral divergence and confusion. Our DDAD operates in a low-dimensional dimensionless domain, enabling robust cross-scene generalization. (a) Single training-detection methods. (b) Original spectral domain cross-scene Methods. (c) Our dimensionless reconstruction cross-scene method.

domain transformation (DDT) module that projects high-dimensional HSIs into unified, source-invariant dimensionless representations for latent subspace reconstruction. This module effectively eliminates spectral divergence across various scenes and mitigates the adverse effects of spectral confusion. Additionally, we propose a controllable dimensionless diffusion reconstruction (CDDR) module to flexibly balance anomaly suppression and background reconstruction through controllable explicit background modeling. To further enhance cross-scene generalization, we design a dynamic spatial–spectral perturbation (DSSP) mechanism to reduce the model's reliance on single-source data characteristics. The contributions of our work are summarized as follows.

1) We propose a novel generalized HAD framework from the perspective of dimensionless domain that effectively eliminates the influence of spectral barriers across scenes in HSIs.
2) We introduce a CDDR module that achieves an optimal balance between background reconstruction and anomaly suppression, further exploring the potential of diffusion models for HAD.
3) We design a DSSP mechanism to reduce the model's dependency on single-source data characteristics.
4) Generalized experiments across four cross-scene HAD datasets demonstrate that DDAD achieves superior generalized anomaly detection performance.

II. Related Works

A. Hyperspectral Anomaly Detection

1) Traditional Methods: Traditional methods can be categorized into two groups: statistics-based methods and representation learning-based methods. Statistics-based methods assume that the background of the HSI obeys some known distribution, statistically obtain the background distribution, and then identify the anomalous pixels by calculating the Mahalanobis distance between the test pixels and the background distribution. The RX detector [2] is the most classical traditional method, which assumes that the background of the HSI obeys a multivariate Gaussian distribution. Inspired by the classical RX detector, a wide variety of variant methods of RX [14], [15], [16], [17], [18], [19] have been proposed afterward contamination of background statistics by anomalies. Statistics-based methods are limited by the assumption of ideal data distribution, and representation learning-based methods have been widely studied. Among them, collaborative representation-based method CRD [3] assumes that background pixels can be reconstructed by linear combinations of other pixels in the spatial neighborhood, while anomalies cannot. The sparse representation-based models [20], [21], [22], [23], [24], [25] reconstruct the background based on a linear combination of background dictionary atoms. Due to the background homogeneity property of HSI, subspace representation-based methods [26] have been proposed for more accurate background modeling. And tensor representation-based methods [27], [28] decompose the HSI into a background tensor and an anomaly tensor, considering both the spectral and spatial dimensions of the tensor. In contrast, structure-aware methods focus on intrinsic data geometry. LSAD [29] leverages local spectral-spatial structure consistency to characterize the background and detect anomalies as deviations from these local patterns. Although traditional methods can achieve promising detection results, they typically rely on manual tuning of image-specific parameters, which substantially hampers their generalization ability across diverse scenes.

2) Deep Learning-Based Methods: Due to the powerful feature learning capabilities of deep learning, deep learning techniques have been widely explored in the field of computer vision [30], [31]. Deep learning-based HAD methods in the presence of unknown prior information usually employ unsupervised learning frameworks. Among them, AE and generative adversarial network (GAN) are widely used in these methods [6], [32], [33], [34], [35], [36], [37], [38], and the use of GAN usually appears in combination with AE. Transformer has been applied to the HAD task due to its ability to better capture the global and local features of the HSI [7], [39], [40]. Recently, several innovative studies have introduced Mamba to the HAD task due to its adaptive input scanning mechanism and the advantage of linear computational complexity [8], [41]. In addition, several cross-image HAD methods have been proposed recently. WeaklyAD [10] employs weakly supervised learning and constructs a reconstruction network based on spectral distance constraints that enhances the discriminative properties of background and anomalies. DFAE [11] adopts frequency-domain learning to detect anomalies in the high-frequency and low-frequency components in parallel, and fuses the two detections as the final result. AET-Net [12] employs a random mask generation strategy to help the reconstruction network learn the contextual relationship between background and anomaly. BSDM [42] adopts the diffusion model to suppress the background of the original HSI, and uses RX for detection. TDD [13] proposes an anomaly sample simulation strategy to generate a large number of pairs of anomalous samples that are used to optimize the model to learn the spectral deviation relation. In recent years, general models [43], [44],

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[45] for remote sensing have become a research hotspot, as a single foundation model can be adapted to multiple tasks with strong generalization capability. UniADRS [43] is a cross-modality anomaly detector that shifts the learning objective from modeling specific background distributions to learning an image-independent consistency deviation metric. UniADRS achieves unified anomaly detection across different remote sensing modalities. *Despite their generalization strengths, they do not effectively resolve the spectral divergence and confusion inherent in HSIs, resulting in suboptimal performance in cross-scene scenarios.*

B. Diffusion Models for Anomaly Detection

Diffusion models [46], [47], [48] have higher generation quality and more diverse coverage patterns compared to AE and GAN. In recent years, diffusion models have been widely used in unsupervised anomaly detection. Due to the powerful denoising and data modeling capabilities and excellent sample generation quality, diffusion models have been applied to normal sample reconstruction or data enhancement in the anomaly detection task and have achieved remarkable results in the fields of industrial defect detection [49], [50], [51], [52] and medical image analysis [53], [54], [55], [56], [57]. However, in the field of hyperspectral imagery, although diffusion models have been widely used in tasks such as denoising [58], image restoration [59], super-resolution [60], classification [61], and change detection [62], etc., *they have not been widely used in anomaly detection.*

III. Method

A. Preliminaries

Denoising diffusion model [46] is probabilistic generative model consisting of two stages: forward diffusion and reverse sample. The forward process is a Markov chain with T timesteps, in which noise obeying a standard Gaussian distribution $\boldsymbol{\epsilon}$ is gradually added to the input image $\mathbf{x}_0$, and the noise-added image $\mathbf{x}_t$ at any time step $t \in [0, T]$ depends only on the previous state $\mathbf{x}_{t-1}$. The noise-added image $\mathbf{x}_t$ can be expressed as follows:

$$\mathbf{x}_t = \sqrt{a_t}\mathbf{x}_{t-1} + \sqrt{1-a_t}\boldsymbol{\epsilon}_{t-1} \tag{1}$$

where a_t is a parameter for the noise scheduler to control the noise intensity added at each time step.

The noise-added image $\mathbf{x}_t$ can be further expressed as (2) related only to $\mathbf{x}_0$ and a_t by derivation, where $\bar{a}_t = \prod_{i=1}^{t} a_i$

$$\mathbf{x}_t = \sqrt{\bar{a}_t}\mathbf{x}_0 + \sqrt{1-\bar{a}_t}\boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{2}$$

The reverse process trains a denoising network to denoise the noise-added image $\mathbf{x}_t$ for obtaining $\mathbf{x}_{t-1}$, and the training objective function can be simplified to predict the noise $\boldsymbol{\epsilon}_t$ added at each time step t, as formulated in the following equation:

$$L = \mathbb{E}_{\mathbf{x}_t, \boldsymbol{\epsilon}}\left[\|\boldsymbol{\epsilon}_t - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|^2\right]. \tag{3}$$

After obtaining the predicted noise $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, the image $\mathbf{x}_{t-1}$ at the previous time step can be generated by inference sampling, as formulated in the following equation:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{a_t}}\left(\mathbf{x}_t - \frac{1-a_t}{\sqrt{1-\bar{a}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right) + \sigma_t \mathbf{z} \tag{4}$$

where σ_t controls the standard deviation of the added noise at each time step, and $\mathbf{z}$ denotes a standard Gaussian noise.

B. DDAD

1) Overview: Fig. 2 illustrates the overall architecture of DDAD framework for cross-scene generalized HAD. DDAD leverages diffusion reconstruction in a dimensionless domain to mitigate spectral divergence and confusion, achieving robust generalization. Given an HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W and B, respectively, denote the height, width, and band number. First, we introduce the DDT module to convert $\mathbf{X}$ into a low-dimensional dimensionless representation $\mathbf{A}_0 \in \mathbb{R}^{H\times W\times L}$, where L denotes the number of channels. During training, the data $\mathbf{A}_0$ from a single dataset is fed into the CDDR module to learn background reconstruction, while a denoising network is trained to reconstruct the background from noise-injected inputs while suppressing the reconstruction of anomalies. Meanwhile, the DSSP mechanism introduces multidimensional perturbations to reduce model's reliance on training data characteristics. During inference, test samples are transformed into dimensionless data $\mathbf{A}$ via the DDT module and fed into the CDDR module to obtain the reconstructed data $\hat{\mathbf{A}}$. Detection results are obtained by calculating the reconstruction error $\sum_{l=0}^{L-1}\|\mathbf{A}_l - \hat{\mathbf{A}}_l\|_2$.

2) Dimensionless Domain Transformation: In the DDAD framework, the transformation of dimensionless domain plays a crucial role in enabling effective cross-scene HAD. Although HSIs contain rich spectral information across their continuous bands, inherent limitations such as spectral divergence, spectral confusion, and high dimensionality significantly hinder the generalization of HAD across different scenes. In real-world scenarios, cross-scene HSIs acquired at different times, locations, or sensors. The spectral discrepancies across images acquired by different sensors mainly arise from variations in spectral response functions and the number of spectral bands, resulting in differing spectral response values and spectral resolution. Additionally, sensors exhibit differing spatial resolution, potentially leading to variations in mixed-pixel composition within images. Furthermore, differences in acquisition location and time introduce varying scenes across images, causing shifts in both background and anomaly spectral distributions.

From our observation, the abundance space serves as an ideal dimensionless domain for cross-scene HAD. First, the abundance space reflects the composition ratio of materials in HSIs, where the abundance maps represent the weighting coefficients of each pixel on each endmember, and these endmembers represent the underlying material components in the image. To address the variation in spectral bands numbers across different HSIs, we transform all HSIs into abundance representations with same channels number. Specifically, abundance maps are extracted using the same number

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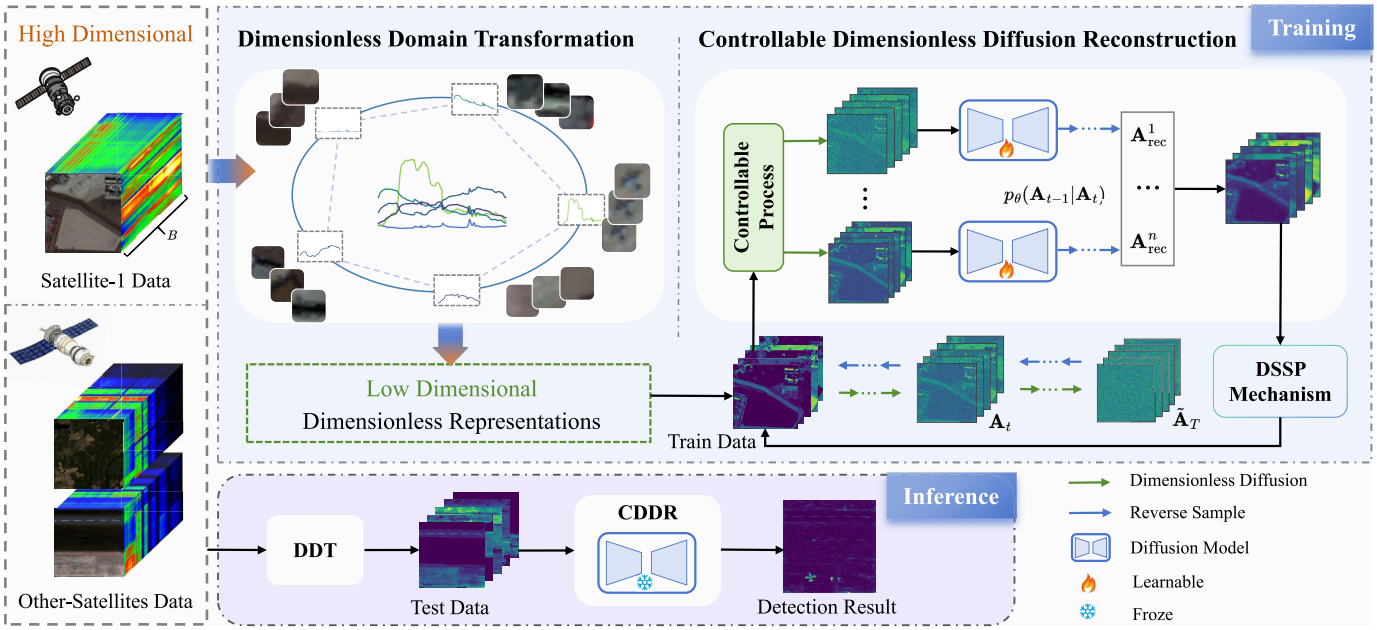


Fig. 2. Overview of DDAD. A DDT module is introduced to convert high-dimensional cross-scene HSIs into low-dimensional dimensionless data for background reconstruction. A CDDR module is employed to reconstruct background from controllable multiscale noisy images. To further enhance generalization, DSSP mechanism is integrated during training to reduce the model's reliance on train data characteristics and improve anomaly suppression.

of endmembers for all datasets, ensuring a consistent feature dimension for the subsequent model input. Additionally, the abundance represent the proportional composition of constituent materials rather than the spectral reflectance values in HSIs, and they are constrained by physical priors, i.e., non-negativity and sum-to-one. As a result, the abundance representation is inherently dimensionless and invariant to sensor-dependent amplitude scaling, making it less sensitive to variations caused by different sensors or acquisition conditions. They capture the intrinsic material structure relationships within a scene and conform to the imbalance property in anomaly detection, where background components dominate and anomalies are sparse. By learning such relative structural representations instead of absolute spectral signatures tied to a specific scene, the model becomes less sensitive to cross-scene spectral distribution shifts, thereby significantly improving generalization performance. This effectively addresses spectral divergence across scenes.

In addition, the abundance space is derived through unmixing, which naturally separates different endmembers and effectively mitigates the issue of spectral confusion. Furthermore, for the HAD task, we are concerned with the ability to effectively distinguish the anomalous pixels from background. Therefore, it is sufficient to only extract abundance maps corresponding to a few representative endmembers, without the need to decompose all materials in the real scene, which effectively transforms the high-dimensional HSIs into low-dimensional dimensionless data

$$\min_{\mathbf{A}^{(i)}\geq 0} \frac{1}{2}\left\|\mathbf{S}^{(i)}\mathbf{A}^{(i)}-\mathbf{X}^{(i)}\right\|_2^2+\lambda\left\|\mathbf{A}^{(i)}\right\|_1, \text{s.t. } \mathbf{1}^T\mathbf{A}^{(i)}=1. \quad (5)$$

Therefore, the abundance space establishes a unified feature domain that inherently normalizes scene-specific variations while preserving essential discriminative features for anomaly detection. Based on this insight, we employ the abundance space as a unified dimensionless domain for generalized HAD. Specifically, given a set of datasets $\mathcal{DS} = \{\mathbf{X}^{(i)}\}_{i=1}^{J}$ from different data sources, J denotes the number of data sources. We choose the vertex component analysis method [63] for endmember extraction to obtain the endmembers $\mathbf{S}^{(i)}$. Subsequently, we use ADMM to solve the optimization problem in (5) to calculate the abundance map $\mathbf{A}^{(i)}$, refer to SUnSAL [64]. In (5), $\lambda > 0$ denotes a regularization parameter.

3) Controllable Dimensionless Diffusion Reconstruction: To achieve a controllable tradeoff between background reconstruction and anomaly suppression, the CDDR module explicitly models the background regions through dimensionless diffusion, while effectively disrupting the reconstruction of anomalous pixels. Specifically, during training, we gradually add random Gaussian noise $\boldsymbol{\epsilon}$ to the abundance map $\mathbf{A}_0$ at each time step in the forward diffusion phase. After T time steps, we transform the input into an approximation of Gaussian noise $\tilde{\mathbf{A}}_T$. Then, in the reverse phase, we train a denoising network to predict the noise $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ at each time step and progressively eliminate it

$$\mathcal{C}(A_0) = \sqrt{\bar{\alpha}_{t_{\text{cond}}}}A_0 + \sqrt{1-\bar{\alpha}_{t_{\text{cond}}}}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon}\sim\mathcal{N}(0,\mathbf{I}) \quad (6)$$

where $\bar{\alpha}_{t_{\text{cond}}}$ represents the cumulative product of the cosine schedule coefficients in the diffusion process. In the context of HAD, our purpose of utilizing the diffusion model is to reconstruct background while effectively suppressing anomalous signatures, rather than generate entirely new data. Therefore, we implement a controllable process $\mathcal{C}(A_0)$ on the input data A_0: we employ a multiscale noise strategy that applies varying noise intensities to effectively handle anomalies of different sizes, thus improving the model's generalization. Specifically, we define an ensemble of time steps $\mathcal{T}_{\text{rec}} = [t_1, t_2, \ldots, t_i \ldots, t_n]$

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that represents different noise levels. Instead of denoising from Gaussian noise, we start from the noise-added data $\mathbf{A}_{t_{\text{cond}}}$ at a controllable time step $t_{\text{cond}} \in \mathcal{T}_{\text{rec}}$ to obtain the reconstructed data. The final reconstruction is obtained through an ensemble averaging process as expressed as follows:

$$\hat{\mathbf{A}} = \frac{1}{n}\sum_{i=1}^{n}\mathbf{A}_{\text{rec}}^{i}$$

$$\text{with } \mathbf{A}_{\text{rec}}^{i} \sim p\left(\mathbf{A}_{t_i}\right)\prod_{t=1}^{t_i} p_\theta\left(\mathbf{A}_{t-1}|\mathbf{A}_t\right) \tag{7}$$

where $\mathbf{A}_{\text{rec}}^{i}$ represents the reconstructed abundance map at the noise level t_i, and p_θ denotes the learned reverse diffusion process. In cross-scene scenarios, this approach not only improves reconstruction stability, but also ensures robust anomaly suppression across various anomaly scales.

4) DSSP Mechanism: To further reduce the model's dependency on the characteristic of train data in cross-scene scenarios and enhance anomaly suppression, we propose a DSSP mechanism. The overall workflow of this mechanism is illustrated in Fig 3. DSSP mechanism is executed dynamically during the model training process to enhance the generalization of the model. Specifically, we first train the model for a limited epochs number p to acquire the basic reconstruction capability. We employ this model for CDDR to generate reconstructed data $\hat{\mathbf{A}}$. Then, to identify potential anomaly locations, we calculate a coarse detection map $\mathbf{D}$ based on the reconstructed data. Subsequently, an adaptive thresholding strategy is applied to generate a binary anomaly mask $\mathbf{M}$, as defined in (8)

$$\mathbf{M}_{i,j} = \begin{cases} 1, & \text{if } \mathbf{D}_{i,j} > S\left(\gamma\right) \\ 0, & \text{otherwise} \end{cases} \tag{8}$$

$$\mathcal{S}\left(\gamma\right) = \text{sorted}\left(\mathbf{d}\right)\left[N - \lfloor \gamma \cdot N \rfloor\right] \tag{9}$$

$\mathcal{S}(\gamma)$ expresses the adaptive score threshold that dynamically adjusts based on the statistical properties of the detection map, where $\mathbf{d}$, N, and γ denote the flattened detection map, the total number of pixels, and the scale factor.

To improve the smoothness and continuity of potential anomaly and background pixel boundaries, we implement a neighborhood-based filling operation to generate the filled map $\mathbf{A}_{\text{fill}}$. This operation replaces potential anomaly regions with the averaged values of their surrounding background pixels to eliminate anomalous signatures while effectively preserving local spatial patterns, as formulated in the following equation:

$$\mathbf{A}_{\text{fill}} = \mathbf{M} \odot \text{F}\left(\mathbf{A} \odot \neg\mathbf{M}\right), \tag{10}$$

where F denotes a 2-D convolution operation employing a kernel of size k, which computes the average value of the neighborhood pixels for each pixel point. $\odot$ denotes the element-wise product. Subsequently, we compute the recombined abundance map $\overline{\mathbf{A}}$ through a weighted combination, as defined in the following equation:

$$\overline{\mathbf{A}} = \mathbf{A}_{\text{fill}} \odot \mathbf{M} + \mathbf{A} \odot \neg\mathbf{M}. \tag{11}$$

Finally, abundance perturbation is employed to enhance the model's generalization capability by breaking its sequential learning bias toward background and anomalies in the abundance space. Given the abundance index sequence $\mathcal{I} = \{1, 2, \ldots, L\}$ corresponding to $\overline{\mathbf{A}}$, we generate a perturbed version via random permutation $\mathcal{P}(r)$ of the sequence, where r denotes the probability of implementing the perturbation. And then we obtain a new sequence $\mathcal{I}'$, producing the perturbed recombined data in the following equation:

$$\mathbf{A}\left[:, :, i\right] = \overline{\mathbf{A}}\left[:, :, \mathcal{I}_i'\right], i \in \{1, 2, \ldots, L\} \tag{12}$$

which is used as input $\mathbf{A}_0$ for the next iteration. To maintain training stability while ensuring effective perturbation, we set a probabilistic execution with $r = 0.5$ to control the execution of abundance perturbations.

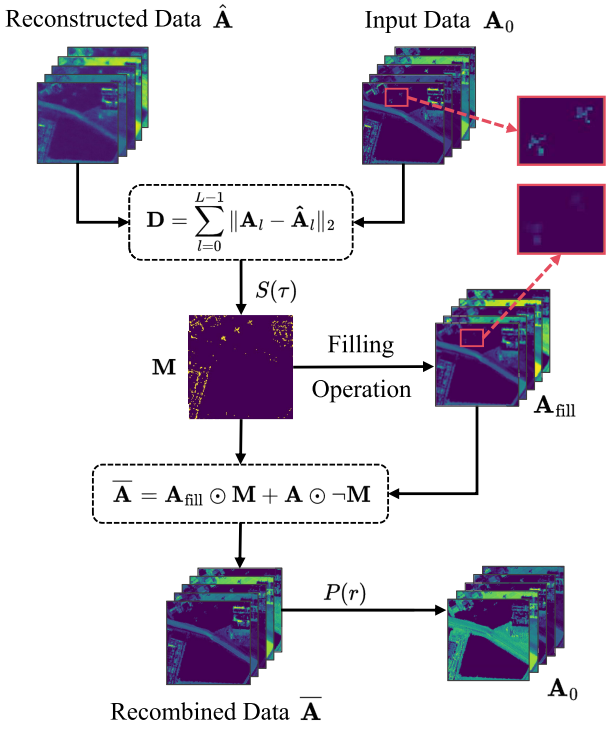


Fig. 3. Illuatration of our DSSP mechanism.

IV. Experiments and Analysis

A. Experimental Settings

1) Datasets: The experimental datasets comprise AVIRIS, Sandiego [23], Gulport [65], and Hyperion [65], which are collected from the airborne visible/infrared imaging spectrometer (AVIRIS) sensor and the EO-1 Hyperion sensor. All datasets used are publicly available. The details of the four datasets from different scenes we used are listed below. And the corresponding pseudo-color maps of each dataset are shown in Fig. 4.

a) AVIRIS: The dataset is captured using the AVIRIS sensor over a coastal area. It consists of 186 spectral bands, with a spatial size of 150 × 150 pixels. The anomalies included in this dataset are three airplanes, which account for 0.32% of the total pixels.

b) Gulfport: The dataset is acquired by the AVIRIS sensor over Gulfport. It comprises 191 spectral bands and has a spatial size of 100 × 100 pixels. The anomalies in this dataset are

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Fig. 4. Pseudo-color maps of four datasets. (a) AVIRIS. (b) Gulfport. (c) San Diego. (d) Hyperion.

three airplanes of different sizes, which account for 0.60% of the total pixels.

c) Sandiego: The dataset is acquired by the AVIRIS sensor over San Diego, California, USA. It includes 189 spectral bands after preprocessing, and has a spatial size of 100×100 pixels. The anomalies in this dataset are also three airplanes, representing 1.34% of the total pixels.

d) Hyperion: The dataset is captured by the EO-1 Hyperion sensor, which provides 155 spectral bands and has a spatial size of 150×150 pixels. The anomalies in this dataset are several storage bins and roofs of small size, which account for approximately 0.08% of the total pixels.

2) Evaluation Metrics: We employ widely used evaluation metrics in HAD: the 2-D and 3-D receiver operating characteristic (ROC) curve, the area under the ROC curve (AUC) score $\text{AUC}_{(P_d,P_f)}$, $\text{AUC}_{(P_d,\tau)}$, $\text{AUC}_{(P_f,\tau)}$ and the box-whisker plots. The ROC curve is a graphical representation of the detection performance, plotting the true positive rate (P_d) against the false positive rate (P_f) under different threshold settings τ. The AUC scores are scalar metric ranging from zero to one used to quantify the overall performance. Specifically, $\text{AUC}_{(P_d,P_f)}$ reflects the overall tradeoff between anomaly detection capability and background suppression, $\text{AUC}_{(P_d,\tau)}$ measures the anomaly response ability, and $\text{AUC}_{(P_f,\tau)}$ characterizes the background false-alarm suppression capability. A 2-D ROC curve closer to the top-left corner with a higher $\text{AUC}_{(P_d,P_f)}$ value indicates better overall detection performance. The box-whisker plots show the background anomaly separation ability of methods, where the red box indicates the distribution of anomalous pixels, and the green box indicates the distribution of background pixels. A greater distance between the red and green boxes indicates a better separation of background and anomaly. The lower height of the green box indicates a better background reconstruction of the method, and the higher red box indicates better anomaly suppression.

3) Implementation Details: All experiments in this article are conducted on a workstation equipped with an NVIDIA RTX 2080Ti GPU, an Intel i9-13900HX CPU and 32GB RAM, using python 3.8, PyTorch 2.0.1, and MATLAB R2024a. In the proposed DDAD framework, the parameters are set as follows: In the DDT module, endmembers number $L = 5$; In the CDDR module, total diffusion time steps $T = 1000$ and fixed time steps $\mathcal{T}_{\text{rec}} = [50, 100, 150, 200, 250, 300]$ for reverse diffusion during training. During cross-scene detection inference, it is adjusted to $[100, 125, 150, 175, 200, 225]$; In the DSSP mechanism, the limited number of epochs $p = 200$, threshold $\gamma = 0.03$, kernel size $k = 3$. The denoising network U-Net is optimized using the AdamW optimizer with a learning rate of 0.0001 and trained for a total of 1000 epochs. The U-Net training process employs the mse loss. A connected component postprocessing filter [66] is applied to the final residual maps to further remove background artifacts.

TABLE I
PARAMETER SETTINGS FOR TRADITIONAL COMPARISON METHODS UNDER DIFFERENT TRAIN DATASET SETTING

Dataset	RX	CRD	LRASR	ADLR	2S-GLRT
	/	$(w_{\text{in}}, w_{\text{out}})$	(K, P, β, λ)	(c, bw, λ)	$(w_{\text{in}}, w_{\text{out}})$
AVIRIS	/	(9, 11)	(19, 20, 0.1, 0.1)	(25, 0.1, 0.1)	(11, 13)
Gulfport		(9, 11)	(13, 20, 0.1, 0.1)	(30, 0.1, 0.1)	(11, 13)
Sandiego		(3, 13)	(13, 20, 0.1, 0.1)	(30, 0.1, 0.1)	(5, 9)
Hyperion		(9, 11)	(19, 20, 0.1, 0.1)	(20, 0.1, 0.1)	(3, 5)

B. Comparison With State-of-the-Art Methods

In order to comprehensively evaluate its performance, we choose eleven comparison algorithms as benchmarks, including five classical traditional detection methods: RX [2], CRD [3], LRASR [23], ADLR [4], and 2S-GLRT [5], as well as six of the current state-of-the-art deep learning-based detection methods: Auto-AD [6], GT-HAD [67], SSHAD [8], AET-Net [12], TDD [13], and UniADRS [43]. Among them, Auto-AD, GT-HAD, and SSHAD are single-image training and detection methods. AET-Net and TDD are state-of-the-art cross-scene detection methods, UniADRS is general anomaly detection method.

To fairly evaluate cross-scene generalization, we adopt a unified parameter tuning strategy for traditional methods: parameters are tuned on a single-source dataset and applied to all test datasets without individual adjustment. The specific parameter settings are summarized in Table I. For deep learning-based methods (except UniADRS), cross-scene experiments are conducted across four datasets. Each method is trained on one dataset, and tested on all four datasets. For single-image training and detection methods (Auto-AD, GT-HAD, SSHAD), we use the band alignment strategy [42] when testing HSIs from different sources. Since UniADRS is a cross-modality anomaly detector rather than a method specifically designed for hyperspectral imagery, we adopt the model trained on the official synthetic dataset provided by the authors and directly evaluate it on the four hyperspectral datasets.

1) Quantitative Results: We quantitatively evaluate the cross-scene detection performance of different methods using AUC scores, and the results are shown in Table II. The best results are in bold, and the second-best results are underlined. Our method achieves robust and excellent cross-scene detection performance on all four datasets. Specifically, the proposed method significantly outperforms the comparison methods in terms of average detection performance on all datasets when trained on different datasets. Compared to deep learning-based cross-scene detectors AET-Net, TDD and general detector UniADRS, our method improves the mean $\text{AUC}_{(P_d,P_f)}$ by 7.33%, 35.58%, and 30.71%, respectively. Additionally, the $\text{AUC}_{(P_d,\tau)}$ and $\text{AUC}_{(P_f,\tau)}$ results demonstrate that our method possesses superior background suppression capabilities and achieves an optimal balance between back-

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TABLE II

AUC SCORES OF DIFFERENT METHODS FOR CROSS-SCENE DETECTION ON FOUR DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, THE SECOND-BEST RESULTS ARE UNDERLINED

Test Dataset	Traditional Method					Single Training-detection Method			Cross-scene Method			
	RX	CRD	LRASR	ADLR	2S-GLRT	Auto-AD	GT-HAD	SSHAD	AET-Net	TDD	UniADRS	**Ours**
Train-AVIRIS												
AVIRIS	0.9237	0.9292	0.9762	0.6333	0.7416	0.9081	0.9644	0.9742	0.9411	0.9832	0.8057	**0.9933**
Gulfport	0.9526	0.9280	0.6764	0.4733	0.9183	0.6491	0.9144	0.5417	0.9023	0.9236	0.4308	**0.9750**
Sandiego	0.9403	0.9415	0.7788	0.5321	0.8282	0.6567	0.9631	0.5656	0.9693	0.8530	0.7793	**0.9833**
Hyperion	0.9978	**0.9982**	0.9793	0.2883	0.6838	0.6456	0.7092	0.2342	0.8552	0.5261	0.6481	0.9767
Average	0.9536	0.9492	0.8527	0.4818	0.7930	0.7149	0.8878	0.5789	0.9170	0.8215	0.6660	**0.9821**
Train-Gulfport												
AVIRIS	0.9237	0.9292	0.9438	0.2965	0.7416	0.1630	0.4542	0.7894	0.8924	0.6007	0.8057	**0.9725**
Gulfport	0.9526	0.9280	0.7502	0.6217	0.9183	0.9813	0.9939	0.6721	0.8600	0.8192	0.4308	**0.9981**
Sandiego	0.9403	0.9415	0.8236	0.5529	0.8282	0.5532	0.8025	0.4806	0.9153	0.4168	0.7793	**0.9425**
Hyperion	0.9978	**0.9982**	0.9752	0.2996	0.6838	0.9154	0.9370	0.2236	0.8650	0.1793	0.6481	0.9688
Average	0.9536	0.9492	0.8732	0.4427	0.7930	0.6532	0.7969	0.5414	0.8832	0.5040	0.6660	**0.9705**
Train-Sandiego												
AVIRIS	0.9237	0.8662	0.9438	0.2965	0.6373	0.4455	0.6691	0.4896	0.8912	0.9361	0.8057	**0.9847**
Gulfport	0.9526	0.8694	0.7502	0.6217	0.8079	0.3219	**0.9595**	0.2272	0.8707	0.4742	0.4308	0.9508
Sandiego	0.9403	0.9490	0.8236	0.5529	0.8467	0.9410	**0.9880**	0.5615	0.9484	0.9142	0.7793	0.9756
Hyperion	0.9978	**0.9979**	0.9752	0.2996	0.7847	0.6697	0.6280	0.7762	0.9045	0.1565	0.6481	0.9734
Average	0.9536	0.9206	0.8732	0.4427	0.7692	0.5945	0.8112	0.5136	0.9037	0.6203	0.6660	**0.9711**
Train-Hyperion												
AVIRIS	0.9237	0.9292	0.9762	0.5317	0.6431	0.1621	0.1185	0.5039	0.9013	0.2325	0.8057	**0.9857**
Gulfport	**0.9526**	0.9280	0.6764	0.4765	0.7831	0.6133	0.2851	0.7282	0.8809	0.2286	0.4308	0.9435
Sandiego	0.9403	0.9415	0.7788	0.5364	0.7657	0.5963	0.5531	0.3814	0.9356	0.6670	0.7793	**0.9515**
Hyperion	0.9978	**0.9982**	0.9793	0.4071	0.8851	0.9973	0.9915	0.8599	0.8640	0.9652	0.6481	0.9934
Average	0.9536	0.9492	0.8527	0.4879	0.7693	0.5923	0.4871	0.6184	0.8955	0.5233	0.6660	**0.9561**
Overall Performance												
$\mathbf{mAUC}_{(P_d,P_f)}\uparrow$	0.9536	0.9421	0.8629	0.4638	0.7811	0.6387	0.7457	0.5631	0.8998	0.6173	0.6660	**0.9731**
$\mathbf{mAUC}_{(P_f,\tau)}\downarrow$	0.0327	0.0538	0.1422	0.0872	0.0373	0.2096	0.1342	0.2157	0.0529	0.1137	0.1208	**0.0129**
$\mathbf{mAUC}_{(P_d,\tau)}\uparrow$	0.1328	0.2334	0.3006	0.0935	0.0781	0.2845	0.2958	0.2461	0.1891	**0.3597**	0.1684	0.2992

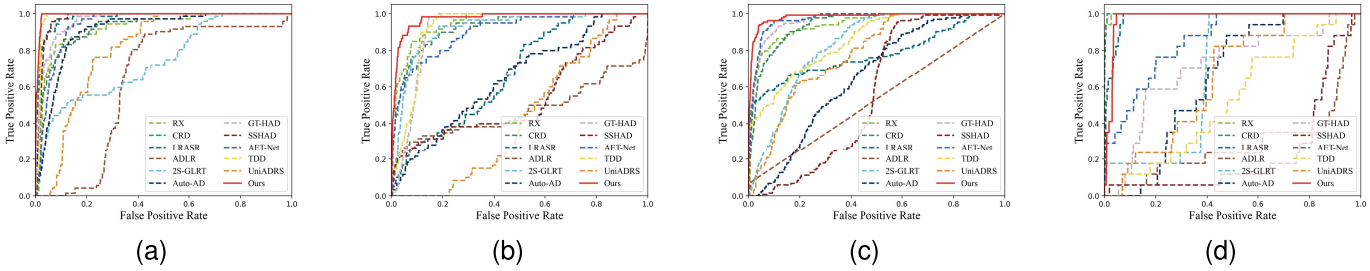


Fig. 5. 2-D ROC curves of different methods on four test dataset. (a) AVIRIS. (b) Gulfport. (c) Sandiego. (d) Hyperion.

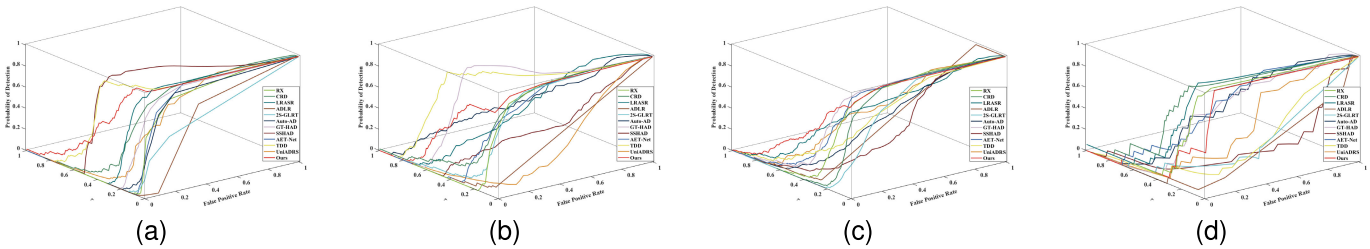


Fig. 6. 3-D ROC curves of different methods on four test dataset. (a) AVIRIS. (b) Gulfport. (c) Sandiego. (d) Hyperion.

ground reconstruction and anomaly suppression. This fully validates the superiority of our method in the cross-scene HAD task.

Figs. 5 and 6 show the 2-D ROC and 3-D ROC curves of the twelve methods in the setting of using AVIRIS as training data and testing across all datasets. As shown in

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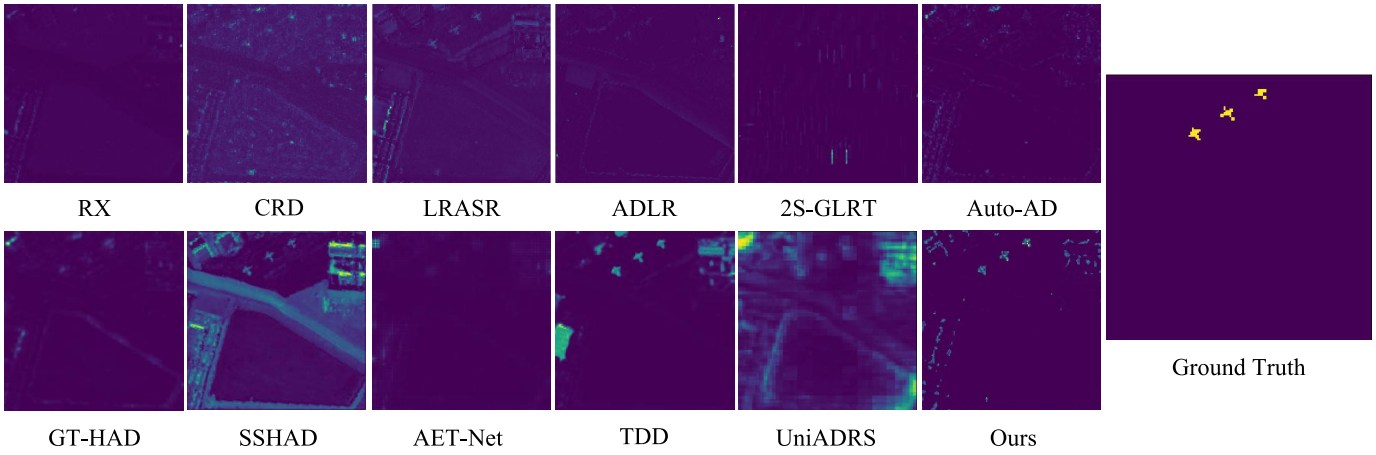


Fig. 7. Anomaly detection maps from different models tested on AVIRIS dataset and the ground truth map of AVIRIS.

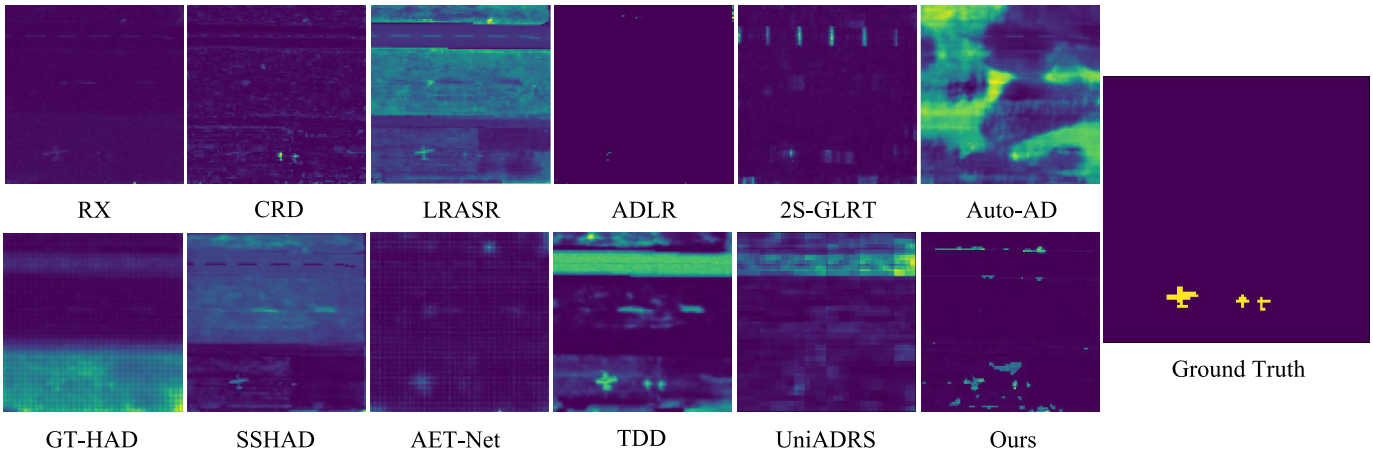


Fig. 8. Anomaly detection maps from different models tested on Gulfport dataset and the ground truth map of Gulfport.

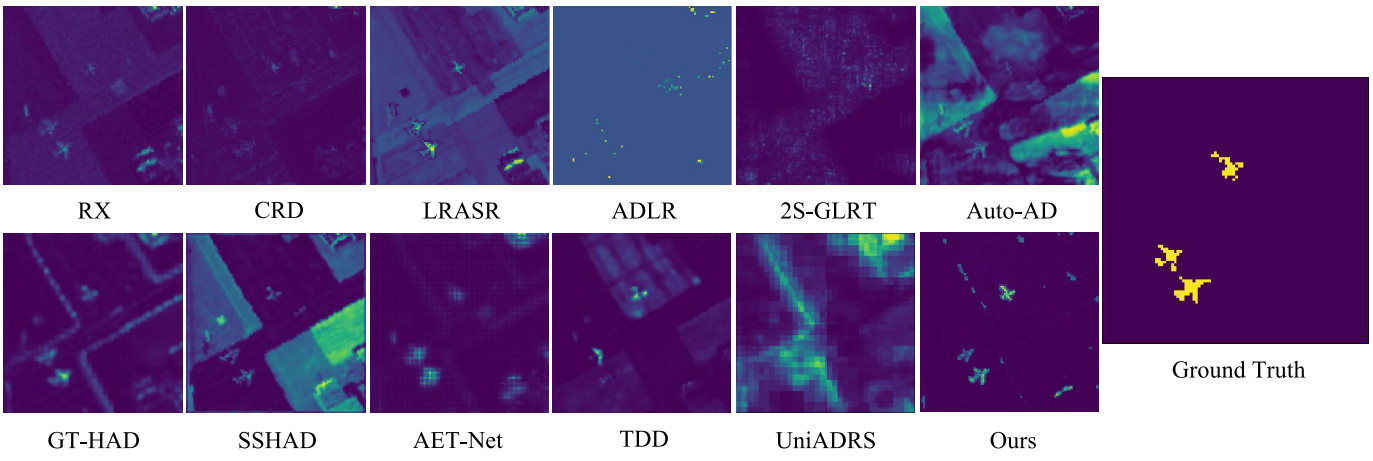


Fig. 9. Anomaly detection maps from different models tested on SanDiego dataset and the ground truth map of SanDiego.

the figures, our method (the red curve) achieves the best performance on most datasets, demonstrating a more robust and stable detection performance in the cross-scene detection settings without large fluctuations in detection accuracy. As shown in the Fig. 6, DDAD presents a balance between the Probability of Detection and the False Positive Rate across different threshold conditions on four datasets, indicating a better balance between background suppression and anomaly response.

2) Qualitative Results: Figs. 7–10 present the detection results of different methods on four datasets, along with the ground truth maps for each dataset, respectively. Our method

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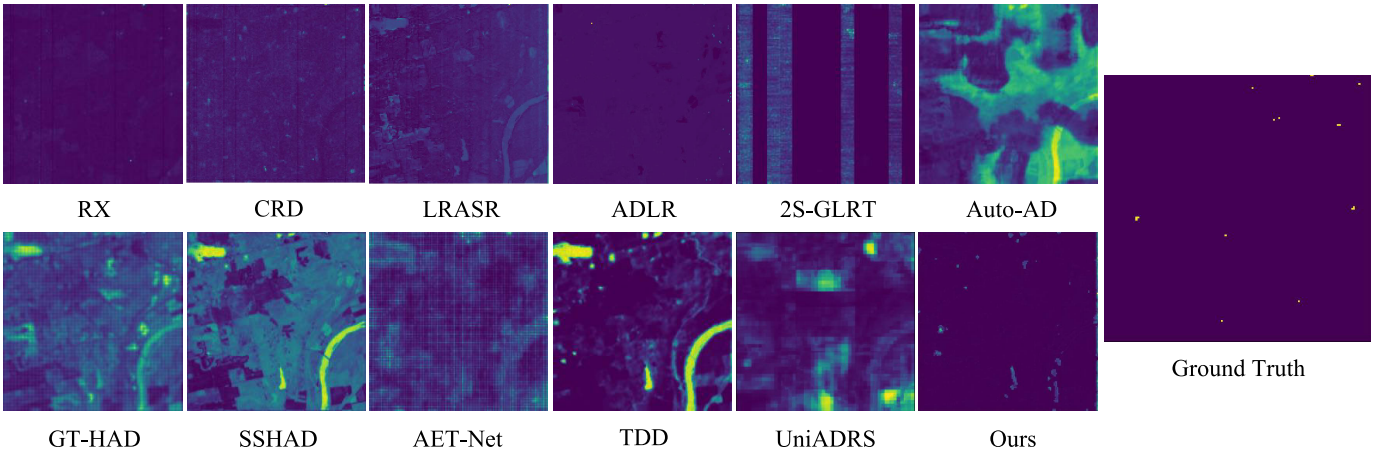


Fig. 10. Anomaly detection maps from different models tested on Hyperion dataset and the ground truth map of Hyperion.

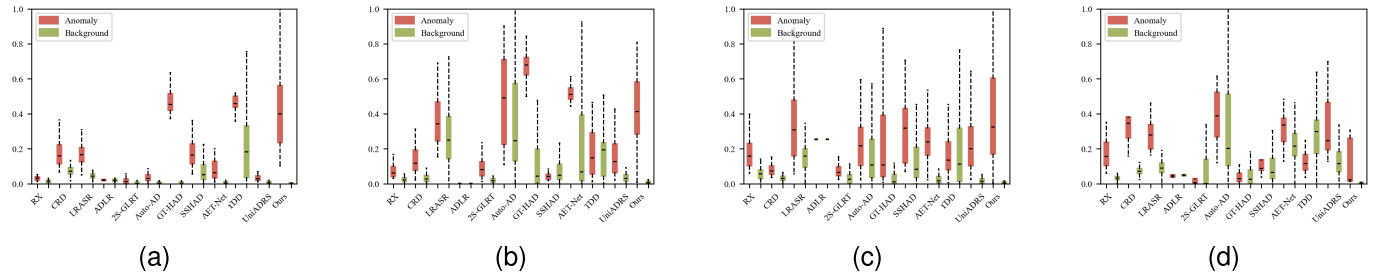


Fig. 11. Box-whisker plots of different methods. AVIRIS is used as the training dataset, while all four datasets are used for inference. (From left to right) Inference datasets correspond to: (a) AVIRIS, (b) Gulfport, (c) Sandiego, and (d) Hyperion.

achieves a superior balance between background suppression and anomaly enhancement. For example, on the AVIRIS dataset as shown in Fig. 7, most traditional methods suppress the background well but also weaken the anomaly signals, while methods like TDD highlight anomalies but introduce high false alarms. In contrast, our method effectively suppresses background interference while preserving anomaly saliency. The deep learning-based comparison methods show unstable cross-scene detection performance, especially on the Hyperion dataset, they produce high false alarms. Our approach demonstrates more stable and accurate cross-scene detection. In addition, our method adaptively detects anomalies of different scales, from large aircraft (in Sandiego) to small objects (in Hyperion).

The box-whisker plots shown in Fig. 11 provide a visual and intuitive understanding of the algorithm's ability to separate anomalies from the background. From the figure, we can see that our method achieves a salient separation of background and anomaly. Despite using the same dataset for both training and testing, our DDAD exhibits substantial improvements over single-image training and detection methods, as well as effective balanced background reconstruction and anomaly suppression on all four test datasets. In cross-scene detection on the other three datasets, DDAD demonstrates superior overall performance, characterized by lower green boxes, higher red boxes, and a greater separation between the two. This result highlights the effectiveness of DDAD in achieving a well-balanced tradeoff between background reconstruction and anomaly suppression.

Fig. 12 presents the anomaly detection results of our proposed DDAD method on four datasets under different training data settings. The results demonstrate that our method consistently achieves stable performance in cross-scene detection. When training and test data originate from the same scene, the results exhibit stronger background suppression and more prominent anomaly enhancement. Even in cross-scene settings, our method maintains an effective balance between background suppression and anomaly highlighting, validating the robustness and generalization capability of the proposed approach.

3) Efficiency Comparison: Since the advantage of generalized methods lies in the ability to process unseen scenes without retraining, we adopt the inference time as the evaluation metric for efficiency comparison. We compare DDAD with the state-of-the-art cross-scene generalized methods AET-Net, TDD and UniADRS. As shown in Table III, DDAD significantly reduces computational cost, demonstrating the efficiency of diffusion in a low-dimensional dimensionless domain. The reported inference time includes the diffusion reconstruction and reconstruction error computation, while the DDT step is performed as a preprocessing step and reused during inference. The average processing time for abundance extraction of DDT is approximately 1.22 s per dataset, which introduces a relatively higher overhead compared to the dif-

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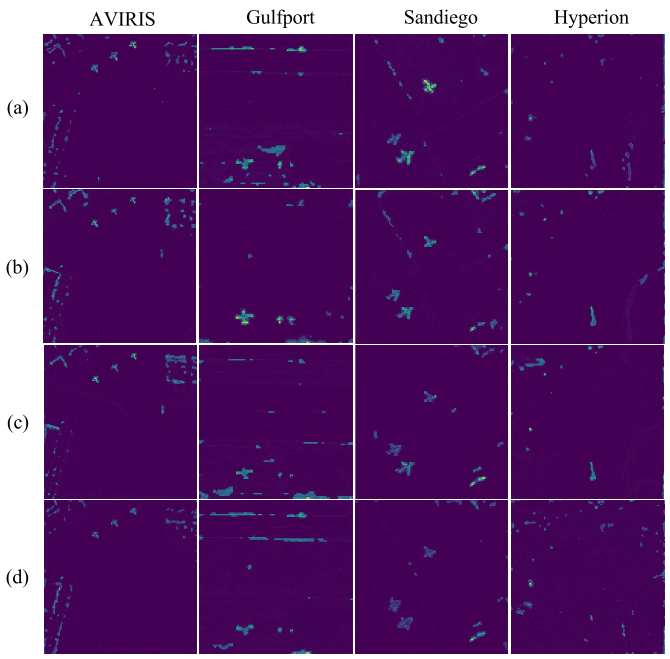


Fig. 12. Detection results of DDAD. (From top to bottom) Train datasets correspond to: (a) AVIRIS, (b) Gulfport, (c) Sandiego, and (d) Hyperion.

TABLE III

EFFICIENCY COMPARISON OF DIFFERENT GENERALIZED METHODS. AVIRIS IS USED AS THE TRAINING DATASET, WHILE ALL FOUR DATASETS ARE USED FOR INFERENCE

Method	AVIRIS	Gulfport	Sandiego	Hyperion	Mean
AET-Net	2.28s	1.77s	0.47s	2.06s	1.65s
TDD	3.55s	3.87s	3.84s	3.54s	3.70s
UniADRS	2.10s	1.48s	1.4s	2.06s	1.76s
DDAD	**0.50s**	**0.58s**	**0.40s**	**0.60s**	**0.52s**

fusion reconstruction stage. However, DDAD still maintains competitive computational efficiency, demonstrating that diffusion in a low-dimensional dimensionless domain enables an effective tradeoff between computational cost and detection performance. Moreover, the inference time of DDAD remains at a practical level, confirming its feasibility for real-world deployment.

C. Ablation Study

To validate the effectiveness of each component in the DDAD framework, we conduct ablation experiments on four datasets. Results are summarized in Table IV. First, we remove the DDT module, and directly use the original HSI as the model input instead of the abundance representation. The cross-scene experimental setting remain unchanged, where the model is trained on a single dataset and evaluated on all datasets. The results show that this variant leads to poor cross-scene performance, which indicates that the spectral divergence and confusion in the original HSIs limit the cross-scene generalization ability. It confirms the effectiveness of our implementation from the perspective of dimensionless domain.

To further illustrate the superiority of dimensionless representations, we show the reconstruction results of the Gulfport dataset using models trained on HSI versus dimensionless

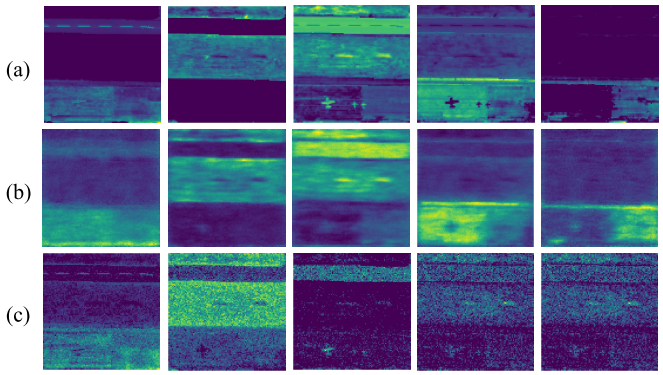


Fig. 13. Comparison of reconstruction results for original HSI and dimensionless data. (a) Raw dimensionless data. (b) Dimensionless data reconstruction result. (c) Original HSI reconstruction result. (From left to right) Flattend abundance maps corresponding to five endmembers.

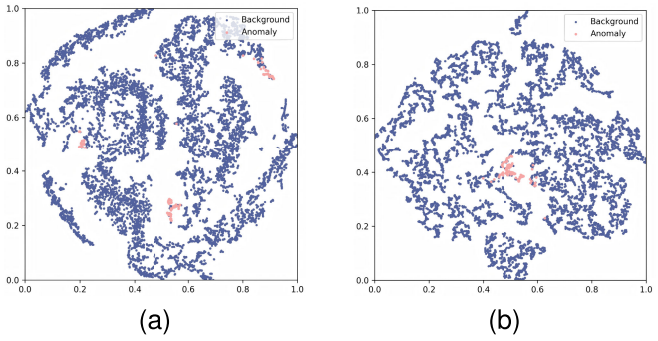


Fig. 14. Feature visualization of (a) high-dimensional original HSI and (b) low-dimensional dimensionless data using t-SNE on Sandiego dataset.

TABLE IV

QUANTITATIVE ABLATION RESULTS OF DIFFERENT METHODS. AVIRIS IS USED AS THE TRAINING DATASET, WHILE ALL FOUR DATASETS ARE USED FOR INFERENCE

Method	AVIRIS	Gulfport	Sandiego	Hyperion	**Average**
DDAD w/o DDT	0.7192	0.5903	0.3620	0.0497	0.4303
DDAD w/o Diff.	0.9102	0.9206	0.5093	0.9262	0.8166
DDAD w/o CDDR	0.9820	0.9475	0.9150	0.9224	0.9417
DDAD w/o DSSP	0.9354	0.9192	0.8932	0.9744	0.9330
DDAD	**0.9933**	**0.9750**	**0.9833**	**0.9767**	**0.9821**

data in Fig. 13. The original result (we transform the reconstructed HSI into the same dimensionless domain for visually comparison) suffers from noise and unsatisfactory background modeling. In contrast, the dimensionless result exhibits superior performance in accurate background reconstruction and effective anomaly suppression. Additionally, Fig. 14 presents the 2-D t-SNE scatter plots of the Sandiego dataset in both the high-dimensional and low-dimensional dimensionless feature spaces. It can be observed that, in the low-dimensional dimensionless space, anomalies in the high-dimensional HSI features exhibit a more sparse and scattered distribution. In contrast, in the dimensionless abundance representation, anomaly samples appear relatively concentrated and separated from the dominant background distribution.

Next, we remove the CDDR module and only use the basic AE framework for background reconstruction (terms as "DDAD w/o Diff."). We can observe that the AE framework shows a significant decrease in AUC scores on all datasets

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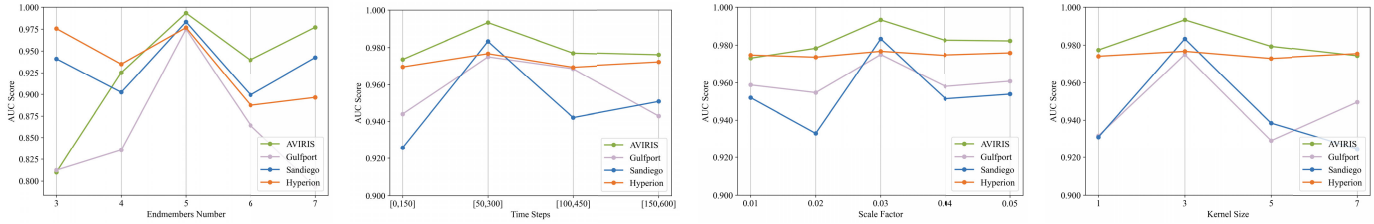


Fig. 15. Parameters analysis of (a) endmembers number L, (b) time steps $\mathcal{T}_{\text{rec}}$, (c) mask computation scale factor τ, and (d) kernel size k. AVIRIS is used as the training dataset, while all four datasets are used for inference.

compared to DDAD, especially the Sandiego dataset. The average AUC score of DDAD leads by 16.55%, demonstrating the effectiveness of the diffusion model for background reconstruction process. To further validate the effectiveness of controllable process in CDDR, we conducted cross-scene detection experiments using a fixed time step of 300 within the same framework(terms as "DDAD w/o CDDR."). Compared with the complete DDAD, the model yields a 4.04% decrease in the average AUC score. Nevertheless, its overall performance still surpasses the AE-based framework by 12.51%, further demonstrating the superiority of the diffusion-based reconstruction model.

Additionally, we remove the DSSP mechanism in the model training process. As can be seen from Table IV, compared to this method, DDAD achieves optimal detection results in all datasets, with an increase of 4.91% in the average AUC scores. It is worth noting that this method has exhibited stable cross-scene detection performance, which also validates the effectiveness of dimensionless diffusion reconstruction in the cross-scene HAD task.

D. Parameter Analysis

We conduct a comprehensive analysis of key hyperparameters in DDAD, including endmembers number L, time steps $\mathcal{T}_{\text{rec}}$, mask computation scale factor τ, and the convolution kernel size k for neighborhood pixel-filling. First, we test the setting of endmembers number $L \in \{3, 4, 5, 6, 7\}$, and the experiment results are shown in Fig. 15(a). We can observe that when $L = 5$, the AUC of the model peaks in all four datasets. It is analyzed that a larger L may overdisperse the background, increasing false positives, while a smaller L may fail to separate anomalies effectively. We evaluate the effect of the time steps $\mathcal{T}_{\text{rec}}$ in CDDR over four ranges: $[0, 150]$, $[50, 300]$, $[100, 450]$, and $[150, 600]$, with six uniformly sampled steps in each range. The results in Fig. 15(b) show that the range [50,300] achieves the best overall performance across four datasets. When T is too small, the injected noise is insufficient to effectively disrupt anomaly structures. Conversely, excessively large T leads to overdiffusion, which degrades background structural information. An intermediate range provides a better tradeoff between background preservation and anomaly suppression, leading to superior detection performance. As observed in Fig. 15(b), for the Hyperion dataset, which contains smaller anomalies with lower proportions and higher spatial resolution, the optimal time steps range remains consistent with other datasets. It also demonstrates that DDAD is robust to variations in spatial resolution and anomaly size distribution. Fig. 15(c) and (d) show the optimal values of γ and k on the four datasets. The impact of different γ and k values on the model performance fluctuates less. We fix $\gamma = 0.03$ and $k = 3$ for overall best performance.

V. Conclusion

In this article, we propose DDAD, a novel generalized HAD framework. Instead of operating in the original spectral domain, DDAD employs a DDT module to map high-dimensional HSIs into a low-dimensional, source-invariant space, effectively eliminating cross-source spectral discrepancies and enhancing feature robustness. To balance background reconstruction and anomaly suppression, we introduce a CDDR module to explicitly model background distributions. To reduce the model's dependency on single-source data characteristics, we design a DSSP mechanism to implement dynamic multidimensional perturbations during training. Extensive experiments on four benchmark datasets demonstrate the strong generalizability of DDAD in cross-scene scenarios.

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